

# Fiber Optics

ICT for Internet and Multimedia

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# Chapter 1

## Basic of EM Waves

### 1.1 Maxwell's Equations

### 1.2 In Time Domain

Electromagnetic phenomena described by **Maxwell equations**.

$$\begin{aligned}\nabla \times \bar{e} &= -\frac{\partial \bar{b}}{\partial t} & \nabla \cdot \bar{d} &= \rho \\ \nabla \times \bar{h} &= \bar{J} + \frac{\partial \bar{d}}{\partial t} & \nabla \cdot \bar{b} &= 0\end{aligned}$$

where:

|                       |    |                                 |                |
|-----------------------|----|---------------------------------|----------------|
| $\bar{e}(\bar{r}, t)$ | is | <b>ELECTRIC FIELD</b>           | $[V/m]$        |
| $\bar{d}(\bar{r}, t)$ | is | the electric flux density       | $[C/m^2]$      |
| $\bar{h}(\bar{r}, t)$ | is | <b>MAGNETIC FIELD</b>           | $[A/m]$        |
| $\bar{b}(\bar{r}, t)$ | is | <b>MAGNETIC FLUX DENSITY</b>    | $[T = Wb/m^2]$ |
| $\rho(\bar{r}, t)$    | is | <b>CHARGE DENSITY</b>           | $[C/m^3]$      |
| $\bar{J}(\bar{r}, t)$ | is | <b>ELECTRIC CURRENT DENSITY</b> | $[A/m^2]$      |
| $\bar{r} = (x, y, z)$ |    |                                 |                |

### 1.3 Algebra Remainder

$$\nabla \text{ is NABLA } \Rightarrow \nabla = \begin{bmatrix} \frac{d}{dx} \\ \frac{d}{dy} \\ \frac{d}{dz} \end{bmatrix}$$

Linear product:

$$\bar{a} \cdot \bar{b} = a_x b_x + a_y b_y + a_z b_z$$

$$\nabla \cdot \bar{d} = \frac{\partial d_x}{\partial x} + \frac{\partial d_y}{\partial y} + \frac{\partial d_z}{\partial z}$$

$\nabla \cdot (\nabla \times \bar{A}) = 0$ , the curl of a vector field is by definition Solenoidal (no divergence)

Same applies when  $\bar{a} \cdot (\bar{a} \times \bar{b}) = 0$  (ORTHOGONAL!)

## 1.4 Harmonic Domain

- In many cases, we can limit the study of EM fields to sinusoidal (or harmonic) signals.
- Convenient to represent fields by their phasor using the so called **complex Steinmetz vectors**.

$$\bar{e}(\bar{r}, t) = \text{Re} [\bar{E}(\bar{r}) \exp(j\omega t)]$$

In this course: **NO ELECTRIC CHARGES** or **ELECTRIC CURRENTS**.

$$\nabla \times \bar{E} = -j\omega \bar{B}$$

$$\nabla \cdot \bar{D} = 0$$

$$\nabla \times \bar{H} = j\omega \bar{D}$$

$$\nabla \cdot \bar{B} = 0$$

## 1.5 Constitutive Relations

The relations between fields and flux densities depend on the medium (mezzo) where the magnetic field is present.

3 media types: **LINEAR, ISOTROPIC, PIECEWISE HOMOGENEOUS**.

The constitutive relations are:

$$\bar{D} = \epsilon \bar{E} \quad \bar{B} = \mu \bar{H}$$

where:

- $\epsilon(\omega)$  is the electric permeability [ $F/m$ ] (Farad/m)
- $\mu(\omega)$  is the magnetic permeability [ $H/m$ ] (Henry/m)

In vacuum: parameters related to speed of light  $\Rightarrow$  constant by definition.

$$c_0 = 299\,792\,458 \text{ m/s}$$

$$\mu_0 = 4\pi \times 10^{-7} \text{ H/m}$$

$$c_0 = \frac{1}{\sqrt{\mu_0 \varepsilon_0}} \Rightarrow \varepsilon_0 = \frac{1}{\mu_0 c_0^2} \approx 8,85 \cdot 10^{-12} \text{ F/m}$$

For generic medium: reflective index defined as  $n = \sqrt{\frac{\mu \varepsilon}{\mu_0 \varepsilon_0}}$ .

## 1.6 Helmholtz Equations

In a linear, isotropic piecewise homogeneous medium without electric charges and currents, Maxwell's equations read:

$$\nabla \times \vec{E} = -j\omega\mu\vec{H}$$

$$\nabla \cdot \vec{E} = 0$$

$$\nabla \times \vec{H} = j\omega\varepsilon\vec{E}$$

$$\nabla \cdot \vec{H} = 0$$

- 1)  $\nabla \times (\nabla \times \vec{E}) = \nabla \times (-j\omega\mu\vec{H})$
- 2)  $\nabla \times (\nabla \times \vec{E}) = -j\omega\mu(\nabla \times \vec{H})$
- 3)  $\nabla \times (\nabla \times \vec{E}) = -j\omega\mu(j\omega\varepsilon\vec{E})$
- 4)  $\nabla \times (\nabla \times \vec{E}) = \omega^2\mu\varepsilon\vec{E}$
- 5)  $\nabla \times \nabla \times \vec{E} = \nabla(\nabla \cdot \vec{E}) - \nabla^2 \vec{E}$  (**Note:**  $\nabla \cdot \vec{E} = 0$ )
- 6)  $\nabla \times \nabla \times \vec{E} = \nabla(0) - \nabla^2 \vec{E} = -\nabla^2 \vec{E}$
- 7)  $-\nabla^2 \vec{E} = \omega^2\mu\varepsilon\vec{E} \Rightarrow \nabla^2 \vec{E} + \omega^2\mu\varepsilon\vec{E} = 0$

$$\nabla^2 \vec{E} + k^2 \vec{E} = 0 \quad \nabla^2 \vec{H} + k^2 \vec{H} = 0$$

where

$$k = \omega\sqrt{\varepsilon\mu}$$